[536709] Stochastic Processes

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4 Continuous time stochastic process

Through this section, we consider $I = \mathbb{R}_+ = [0, \infty)$ which is continuous time. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We define **stochastic process** as a collection of random variables on measurable space. In general, let S be a topological space (or compact metric space) and $\mathcal{B} = \mathcal{B}(S)$ be a Borel set which is a σ-algebra generated by the family of open sets in S.

Stochastic process To describe the d-dimensional system randomly deviating along the time, we denote by $X_t(\omega)$ the realization at time t. If we fix $t \in I$, then $X_t(\cdot)$ is a d-dimensional random variable on Ω, and if we fix ω, then $X_\cdot(\omega)$ is a d-dimensional function on I, which is called **path**. We recall the definition of stochastic process.

Definition 4.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $(S, \mathcal{B})$ be a measurable space and $(X_t)_{t\in I}$ be a collection of random variables

$$X_t \colon \Omega \to S \tag{4.1}$$

Then $\mathbf{X} = (X_t)_{t\in I}$ is called **stochastic process** and $(S, \mathcal{B})$ is called **state space** when

- for any $t \in I$, X_t is d-dimensional random variable
- for any $t \in I$,
$$\mathbb{E}[|X_t|^2] < \infty, \qquad \lim_{s\to t} \mathbb{E}[|X_t - X_s|^2] = 0 \tag{4.2}$$
- for almost every ω, $X_\cdot(\omega)$ called **path** is a discoutinuous function of the first kind.

The condition $\mathbb{E}[|X_t|^2] < \infty$ yields that the mean and covariance are well defined, also guarantees exchangeable limit lim and expectation $\mathbb{E}$, which tells us that stochastic process is the limit of

"good" functions. The second condition

$$\lim_{s\to t}\mathbb{E}[|X_t - X_s|^2] = 0 \tag{4.3}$$

is L^2-continuity, which shows the trajectories do not have big jumps and are relatively smooth. Therefore, even if the process has deviations, it is possible to treat the analysis of the behavior of stochastic processes by calculus.

4.1 Modification

In probability theory two random variables ξ and η on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$ are identified when

$$\mathbb{P}(\xi = \eta) = 1 \tag{4.4}$$

Similarly two stochastic processes $\mathbf{X}$ and $\mathbf{Y}$ are identified when

$$\mathbb{P}(\forall t \in I,\, X_t(\omega) = Y_t(\omega)) = 1, \tag{4.5}$$

which is called *indistinguishable*. When the processes $\mathbf{X}$ and $\mathbf{Y}$ are indistinguishable, it shows the uniqueness in the level of sample path. A weaker condition

$$\mathbb{P}(X_t(\omega) = Y_t(\omega)) = 1 \qquad \forall t \in I \tag{4.6}$$

yields that $\mathbf{Y}$ is a **modification** of $X(\omega)$ ($\mathbf{X}$ is a modification of $\mathbf{Y}$). Much weaker condition is called *equality in law*:

$$\mathbf{X} \overset{\mathrm{d}}{=} \mathbf{Y}, \tag{4.7}$$

that is, for any $n \in \mathbb{N}$ and $t_1, t_2, \cdots, t_n$,

$$(X_{t_1}, X_{t_2}, \cdots, X_{t_n}) \overset{\mathrm{d}}{=} (Y_{t_1}, Y_{t_2}, \cdots, Y_{t_n}). \tag{4.8}$$

Example 4.2. Let $\Omega = [0,1]$ and $I = [0,1]$. We consider

$$X_t(\omega) = \mathbb{1}_{\{t=\omega\}} \tag{4.9}$$

and $Y_t(\omega) = 0$ on $[0,1] \times [0,1]$. Then

$$\{\omega \in \Omega \colon X_t(\omega) = Y_t(\omega)\} \tag{4.10}$$

has Lebesgue measure 1, which implies $Y(\omega)$ is a modification of $X(\omega)$. However, since for all t, there is no ω such that $X_t(\omega) = Y_t(\omega)$, $\mathbf{X}$ and $\mathbf{Y}$ cannot be identified.

When I is countable, then (4.5) and (4.6) are equivalent. if the function is discontinuous of the first kind, then (4.5) and (4.6) are also equivalent. Therefore, to define continuous stochastic process, we assume that for almost every ω path $X_\cdot(\omega)$ is a discontinuous function of the first kind on I. Of course, if we consider "continuous" stochastic process, then we do not have to be so nervous about this point.

Identification

- **Equality in law**: *existence* of stochastic process following a given distribution

  $$\mathbb{P}(\forall t \in I,\, X_t(\omega) = Y_t(\omega)) = 1, \tag{4.11}$$

  $\rightarrow$ **Kolmogorov extension theorem**

- **Modification**: *continuity* of stochastic process defined by Kolmogorov extension theorem

  $$\mathbb{P}(X_t(\omega) = Y_t(\omega)) = 1 \qquad \forall t \in I \tag{4.12}$$

  $\rightarrow$ **Kolmogorov modification theorem**

- **Indistinguishable**: *uniqueness* in the sample path level

  $$\mathbf{X} \overset{\mathrm{d}}{=} \mathbf{Y}, \tag{4.13}$$

We note here that if $\mathbf{X}$ and $\mathbf{Y}$ are right(left)-continuous and $\mathbf{X}$ is a modification of $\mathbf{Y}$, then they are indistinguishable.

Kolmogorov extension theorem The theorem guarantees the unique existence of infinite dimensional stochastic process if arbitrary set of finite dimensional distributions satisfies "*consistency*" condition:

- symmetry: joint distribution for (t_1, t_2) is the same as that for (t_2, t_1)
- marginality: joint distribution integrated over t_3 for (t_1, t_2, t_3) is the same as that for (t_1, t_2)

Let Π_n be a projection from $\mathbb{R}^\infty$ to $\mathbb{R}^n$, that is, for $x = (x_1, x_2, \cdots, x_n, \cdots)$,

$$\Pi_n x = (x_1, x_2, \cdots, x_n) \tag{4.14}$$

Set a σ-algebra $\mathcal{B}(\mathbb{R}^\infty) = \sigma(\Pi_n^{-1} A\colon\ A \in \mathcal{B}(\mathbb{R}^n),\ n = 1, 2, \cdots)$.

Theorem 4.3 (Kolmogorov extension theorem I)**.** *If the probability μ_n $(n = 1, 2, \cdots)$ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ satisfies*

$$\mu_{n+1}(A \times \mathbb{R}) = \mu_n(A), \qquad A \in \mathcal{B}(\mathbb{R}^n), \tag{4.15}$$

then there exists the unique probability μ_∞ on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that

$$\mu_\infty(\Pi_n^{-1} A) = \mu_n(A) \qquad A \in \mathcal{B}(\mathbb{R}^n), n = 1, 2, \cdots \tag{4.16}$$

We have generalized version. Let $\mathbf{T} = [0, T]$ $(T \in \mathbb{R}_+)$ and $\mathcal{T}$ be the family of all countable sets $\mathbf{U}$ contained in $\mathbf{T}$. We denote by $\mathbb{R}^{\mathbf{T}}$the set of all real-valued functions on $\mathbf{T}$ and by $\Pi_{\mathbf{U}}$ the projection from $\mathbb{R}^{\mathbf{T}}$ to $\mathbb{R}^{\mathbf{U}}$. For instance, if $\mathbf{U} = (t_1, t_2, \cdots, t_n)$, then $\mathbb{R}^{\mathbf{U}} = \mathbb{R}^n$ and for function f on $\mathbf{T}$,

$$\Pi_{\mathbf{U}} f = (f(t_1), f(t_2), \cdots, f(t_n)) \tag{4.17}$$

and also for $A \in \mathcal{B}(\mathbf{R}^n)$,

$$\Pi_{\mathbf{U}}^{-1} A = \{f \in \mathbb{R}^{\mathbf{T}}\colon\ (f(t_1), f(t_2), \cdots, f(t_n)) \in A\} \tag{4.18}$$

Set a σ-algebra $\mathcal{B}(\mathbb{R}^{\mathbf{T}}) = \sigma(\Pi_{\mathbf{U}}^{-1} A\colon\ A \in \mathcal{B}(\mathbb{R}^{\mathbf{U}}),\ \mathbf{U} \in \mathcal{T})$.

Theorem 4.4 (Kolmogorov extension theorem II)**.** *For $\mathbf{U}, \mathbf{V} \in \mathcal{T}$, when $\mathbf{U} \subset \mathbf{V}$, we define $\Pi_{\mathbf{U}}^{\mathbf{V}}$ a projection from $\mathbb{R}^{\mathbf{V}}$ to $\mathbb{R}^{\mathbf{U}}$. For all $\mathbf{U} \in \mathcal{T}$, if the probability $\mu_{\mathbf{U}}$ on $(\mathbb{R}^{\mathbf{U}}, \mathcal{B}(\mathbb{R}^{\mathbf{U}}))$ satisfies*

$$\mu_{\mathbf{V}}((\Pi_{\mathbf{U}}^{\mathbf{V}})^{-1} A) = \mu_{\mathbf{U}}(A), \qquad A \in \mathcal{B}(\mathbb{R}^{\mathbf{U}}), \tag{4.19}$$

then there exists the unique probability μ on $(\mathbb{R}^{\mathbf{T}}, \mathcal{B}(\mathbb{R}^{\mathbf{T}}))$ such that

$$\mu(\Pi_{\mathbf{U}}^{-1} A) = \mu_{\mathbf{U}}(A) \qquad A \in \mathcal{B}(\mathbb{R}^{\mathbf{U}}) \tag{4.20}$$

4.2 Continuous stochastic processes

Definition 4.5. The process $\mathbf{X} = \{X_t\}_{t \in I}$ is called a **continuous stochastic process** when the almost all paths are continuous functions on I.

We give a sufficient condition for continuous stochastic process in one-dimension.

Proposition 4.6 (Kolmogorov modification theorem)**.** *For any $s, t \in I$, if there exist positive constants L, δ, γ such that*

$$\mathbb{E}[|X_t - X_s|^\gamma] \leq L\,|t - s|^{1+\delta}, \tag{4.21}$$

then $\mathbf{X}$ is a continuous stochastic process (there exists a modification that has a.s.-continuous sample path for $\mathbf{X}$).

Proof. For simplicity we take $I = [0, 1]$. We fix α such that

$$0 < \alpha < \frac{\delta}{\gamma} \tag{4.22}$$

For dyadic decomposition $\Delta_n(k, \omega) := X_{2^{-n}\,k}(\omega) - X_{2^{-n}\,(k-1)}(\omega)$ $(k = 1, 2, \cdots 2^n)$, by Chebyshev's inequality and the assumption, we have

$$\begin{aligned} \mathbb{P}\left(\max_k |\Delta_n(k,\omega)| > 2^{-n\alpha}\right) &\le \sum_{k=1}^{2^n} \mathbb{P}(|\Delta_n(k,\omega)| > 2^{-n\alpha}) \\ &\le \sum_{k=1}^{2^n} \mathbb{E}[|\Delta_n(k,\omega)|^\gamma]\, 2^{n\alpha\gamma} \\ &\le 2^{-n(\delta-\alpha\gamma)}\, L \end{aligned} \tag{4.23}$$

By the Borel–Cantelli lemma, we can take Ω_0 with $\mathbb{P}(\Omega_0) = 1$ satisfying the following: For any $\omega \in \Omega_0$, there is $n_0 = n_0(\omega)$ such that for any $n \ge n_0$,

$$\max_k |\Delta_n(k, \omega)| \le 2^{-n\alpha} \tag{4.24}$$

Thus for the binary rational point $t = 2^{-n}\, k + \sum_{p=1}^{\ell} 2^{-n+p}\theta_p$ $(\theta_p \in (0, 1))$ in the interval $[2^{-n}\, k, 2^{-n}\, (k + 1))$, and $n \ge n_0$,

$$|X_{2^{-n}\,k}(\omega) - X_t(\omega)| \le \sum_{p=1}^{\ell} \max_j |\Delta_{n+p}(j, \omega)| \le 2^{-n\alpha} \sum_{p=1}^{\ell} 2^{-p\alpha} \le 2^{-n\alpha}(1 - 2^{-\alpha n})^{-1} \tag{4.25}$$

When $|t - s| < 2^{-n}$, we have

$$|X_t(\omega) - X_s(\omega)| \le 3 \sum_{j=n+1}^{\infty} 2^{-j\alpha} = \frac{3 \cdot 2^{-(n+1)\alpha}}{1 - 2^{-\alpha}}, \tag{4.26}$$

which implies that $X_\cdot(\omega)$ is uniformly continuous on dyadic rationals, and thus can be uniquely extended to a continuous path on I. Since dyadic rationals D is dense in I and $X_\cdot(\omega)$ is uniformly continuous on D for $\omega \in \Omega_0$, there exists a unique continuous extension $\widetilde{X}_\cdot(\omega)$ to the entire interval I. For $\omega \notin \Omega_0$, we can simply set $\widetilde{X}_t(\omega) = 0$. Furthermore, the L^γ-convergence assumption implies that for each t, $X_{s_n} \to X_t$ in probability as $s_n \to t$ $(s_n \in D)$. Combined with the construction of $\widetilde{X}$, this ensures that

$$\mathbb{P}(X_t = \widetilde{X}_t) = 1 \qquad \forall t \in I. \tag{4.27}$$

Thus $\widetilde{X}$ is a continuous modification of X. Consequently, the paths are α-Hölder continuous. ■

The equation (4.26) implies the following continuity: for $\varepsilon > 0$, there is $n_1 = n_1(\varepsilon)$ depending only on L, γ, δ such that for any $n \ge n_1$,

$$\mathbb{P}\left(\sup_{|t-s|<2^{-n}} |X_t - X_s| > \frac{3c}{c-1}\, c^{-n}\right) < \varepsilon \tag{4.28}$$

where $c = 2^{\frac{\delta}{2\gamma}}$.

Example 4.7 (Brownian motion)**.** Let ξ_n $(n = 0, 1, \cdots)$ be an independent random variables identically distributed as $\mathcal{N}(0, 1)$. We set

$$X_t := \frac{t}{\sqrt{\pi}}\, \xi_0 + \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \frac{\sin(nt)}{n}\, \xi_n \qquad t \in [0, \pi] \tag{4.29}$$

Then since $(X_{t_1}, X_{t_2}, \cdots, X_{t_\ell})$ is a ℓ-dimensional Gaussian random variable, $\mathbf{X}$ is a Gaussian process with 0 mean and covariance is given by

$$\Gamma(s,t) := \mathrm{Cov}(X_s, X_t) = \frac{st}{\pi} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(ns)\ \sin(nt)}{n^2} = \min(s,t) \tag{4.30}$$

Indeed, fix s and set $f(t) = \min(s,t)$ $(t \in [0,\pi])$. Since $f(0) = 0$, we can extend as

$$f(t) = \sum_{n=1}^{\infty} a_n \sin(nt) \tag{4.31}$$

Computing a_n, we can obtain

$$f(t) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(ns)\ \sin(nt)}{n^2} - \frac{2s}{\pi}\sum_{n=1}^{\infty}\frac{\cos(n\pi)\ \sin(nt)}{n} \tag{4.32}$$

On the other hand, we consider the Fourier expansion of $h(t) = t$ and then obtain

$$t = -2\sum_{n=1}^{\infty}\frac{\sin(n\pi)\ \sin(nt)}{n} \tag{4.33}$$

which shows (4.30). For $\theta < s < t$,

$$\mathrm{Cov}(X_t - X_s, X_\theta) = \Gamma(t,\theta) - \Gamma(s,\theta) = 0 \tag{4.34}$$

Therefore, by Proposition 4.25, the increment $X_t - X_s$ is independent of $\{X_\theta\}_{\theta \le s}$, and since

$$\mathbb{E}[|X_t - X_s|^4] = 3|t-s|^2 \tag{4.35}$$

$\mathbf{X}$ is a continuous stochastic process. The process $\mathbf{X}$ is called **Brownian motion** on $[0,\pi]$.

Next we construct the Brownian motion on $[0,\infty)$. Let $\mathbf{X}^{(n)}$ $(n = 0,1,2,\cdots)$ be independent Brownian motions on $[0,\pi]$. We connect those Brownian motion and make the Brownian motion on $[0,\infty)$:

$$B_t := \begin{cases} X_t^{(0)} & 0 \le t \le \pi, \\ X_\pi^{(0)} + X_{t-\pi}^{(1)} & \pi < t \le 2\pi, \\ \cdots & \\ X_\pi^{(0)} + \cdots + X_\pi^{(k-1)} + X_{t-k\pi}^{(k)} & k\pi < t \le (k+1)\pi. \end{cases} \tag{4.36}$$

Then B_t is a Gaussian process satisfying

$$\mathbb{E}[B_t] = 0, \qquad \mathrm{Cov}(B_t, B_s) = \min(t,s) \tag{4.37}$$

and $B_t - B_s$ is independent of $\{B_\theta\}_{\theta \le s}$. □

Distribution of the continuous stochastic process We consider the distribution of the path. First we set the space of d-dimensional continuous functions on I denoted by $C = C(I;\mathbb{R}^d)$. For continuous stochastic processes $\mathbf{X}$, since the path $X(\omega) = X(\cdot,\omega) \in C$, $\mathbf{X}$ is a C-valued random variable on probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We denote by $\mu_{\mathbf{X}}$ the distribution $\{X_t\}_{t\in[0,T]}$.

for $f \in C$, $\pi_{t_1,t_2,\cdots,t_n} f := (f(t_1), f(t_2), \cdots, f(t_n))$. Set

$$\pi_{t_1,t_2,\cdots,t_n}^{-1} A = \{f \in C\colon (f(t_1), f(t_2), \cdots, f(t_n)) \in A\} \tag{4.38}$$

Then $f \in \pi_{t_1,t_2,\cdots,t_n}^{-1} A$ is equivalent to $(f(t_1), f(t_2), \cdots, f(t_n)) \in A$. Therefore,

$$(X(\omega) \in \pi_{t_1,t_2,\cdots,t_n}^{-1} A) = ((X_{t_1}(\omega), X_{t_2}(\omega), \cdots, X_{t_n}(\omega)) \in A) \tag{4.39}$$

For $n \in \mathbb{N}$, $t_1, t_2, \cdots, t_n \in I$ and $A \in \mathcal{B}^{dn}$, we denote by $\mathcal{B}$ σ-algebra on C generated by

$$\{\pi^{-1}_{t_1,t_2,\cdots,t_n} A \colon A \in \mathcal{B}^{dn},\, t_1, t_2, \cdots, t_n \in I,\, n \in \mathbb{N}\} \tag{4.40}$$

Thus since

$$\mathbb{P}((X_{t_1}(\omega), X_{t_2}(\omega), \cdots, X_{t_n}(\omega)) \in A) = \mathbb{P}(\mathbf{X} \in \pi^{-1}_{t_1,t_2,\cdots,t_n} A), \tag{4.41}$$

for any n, $t_1, t_2, \cdots, t_n$ and A, we define

$$\mu_{\mathbf{X}}(\pi^{-1}_{t_1,t_2,\cdots,t_n} A) = \mathbb{P}((\mathbf{X} \in \pi^{-1}_{t_1,t_2,\cdots,t_n} A). \tag{4.42}$$

By Kolmogorov extension theorem, the probability on C satisfying (4.42) exists uniquely. Thus we denote by $\mu_{\mathbf{X}}$, which is called **distribution** or **law** of $\mathbf{X}$. From (4.42), we have the following theorem.

Theorem 4.8. *If the continuous stochastic processes* $\mathbf{X}$ *and* $\mathbf{Y}$ *have the same finite dimensional distribution, then*

$$\mu_{\mathbf{X}} = \mu_{\mathbf{Y}} \tag{4.43}$$

We note that the continuous stochastic processes $\mathbf{X}$ and $\mathbf{Y}$ have the same finite dimensional distribution when for any n, $t_1, t_2, \cdots, t_n$ and A,

$$\mathbb{P}((X_{t_1}(\omega), X_{t_2}(\omega), \cdots, X_{t_n}(\omega)) \in A) = \mathbb{P}((Y_{t_1}(\omega), Y_{t_2}(\omega), \cdots, Y_{t_n}(\omega)) \in A) \tag{4.44}$$

Example 4.9 (Brownian bridge)**.** Let $\mathbf{X}$ be a Brownian motion. Set

$$\xi(t) = X_t - tX_1 \qquad t \in [0,1] \tag{4.45}$$

Then $\xi(0) = \xi(1) = 0$ and $\xi = (\xi(t))_{t\in[0,1]}$ is a continuous Gaussian process with $\mathbb{E}[\xi(t)] = 0$. For $0 < s < t < 1$,

$$\mathbb{E}[\xi(s)\,\xi(t)] = s(1-t) \tag{4.46}$$

On (t,x)-plane, $(t, \xi(t))$ is called **Brownian bridge**, which is an interpolation between $(0,0)$ and $(1,0)$ by Brownian motion. If we set

$$\eta(t) = (1+t)\,\xi\left(\frac{1}{1+t}\right) \qquad t \in [0,\pi] \tag{4.47}$$

then $\eta = (\eta(t))_{t\in[0\pi]}$ is also a continuous Gaussian process. Moreover, since

$$\mathbb{E}[\eta(t)] = 0, \quad \mathbb{E}[\eta(s)\eta(t)] = (1+t)(1+s)\mathbb{E}\left[\xi\left(\frac{1}{1+s}\right)\,\xi\left(\frac{1}{1+t}\right)\right] = \min(s,t), \tag{4.48}$$

η has the same finite dimensional distribution. Therefore, η is a Brownian motion. □

4.2.1 Convergence of continuous stochastic processes

For d-dimensional continuous function $f = (f_1, f_2, \cdots, f_d)$, we call

$$\|f\| = \sup_{t\le T} \max_{j=1,2,\cdots,d} |f_j(t)| \tag{4.49}$$

norm of the function f. We say that f_n converges to f if

$$\lim_{n\to\infty} \|f_n - f\| = 0. \tag{4.50}$$

We consider the space of continuous functions with norm $\|\cdot\|$ denoted by C. For a real-valued function F on C, we say that F is continuous on $f \in C$ if

$$f_n \to f \quad \Rightarrow \quad F(f_n) \to F(f). \tag{4.51}$$

If F is continuous on all $f \in C$, then we simply say that F is continuous. Since the continuous stochastic processes are C-valued random variables, we define the several modes of convergence.

Definition 4.10 (Convergences almost surely, in probability and in law)**. (i) $\mathbf{X}_n$ converges to $\mathbf{X}$ almost surely** if for almost all ω,

$$\lim_{n\to\infty} \|X_n(\omega) - X(\omega)\| = 0 \tag{4.52}$$

and we denote by $\mathbf{X}_n \overset{\text{a.s.}}{\to} \mathbf{X}$.

(ii) $\mathbf{X}_n$ converges to $\mathbf{X}$ in probability if for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(\|X_n(\omega) - X(\omega)\| > \varepsilon\big) = 0 \tag{4.53}$$

and we denote by $\mathbf{X}_n \overset{\text{pr}}{\to} \mathbf{X}$.

(iii) $\mathbf{X}_n$ converges to $\mathbf{X}$ in law if for any bounded continuous function $F \in C$,

$$\lim_{n\to\infty} \mathbb{E}[F(\mathbf{X}_n)] = \mathbb{E}[F(\mathbf{X})] \tag{4.54}$$

and we denote by $\mathbf{X}_n \overset{\text{d}}{\to} \mathbf{X}$.

Let μ_n and μ be the distribution of $\mathbf{X}_n$ and $\mathbf{X}$, respectively. Then since

$$\mathbb{E}[F(\mathbf{X}_n)] = \int_C F(h)\ \mu_n(dh), \qquad \mathbf{E}[F(\mathbf{X})] = \int_C F(h)\ \mu(dh), \tag{4.55}$$

the convergence in law implies the weak convergence $\mu_n \to \mu$.

Proposition 4.11.

$$\mathbf{X}_n \overset{pr}{\to} \mathbf{X} \ \text{ and } \ \mathbf{X}_n \overset{pr}{\to} \mathbf{Y} \quad \Rightarrow \quad \mathbb{P}(\mathbf{X} = \mathbf{Y}) = 1. \tag{4.56}$$

Proof. Since

$$\|X(\omega) - Y(\omega)\| \le \|X(\omega) - X_n(\omega)\| + \|X_n(\omega) - Y(\omega)\|, \tag{4.57}$$

we have

$$\mathbb{P}(\|\mathbf{X} - \mathbf{Y}\| > 2\varepsilon) \le \mathbb{P}(\|\mathbf{X} - \mathbf{X}_n\| > \varepsilon) + \mathbb{P}(\|\mathbf{X}_n - \mathbf{Y}\| > \varepsilon) \tag{4.58}$$

As $n \to \infty$, we obtain that $\mathbb{P}(\|\mathbf{X} - \mathbf{Y}\| > 2\varepsilon) = 0$, which completes the proof. ■

Proposition 4.12.

$$\mathbf{X}_n \overset{law}{\to} \mathbf{X} \ \text{ and } \ \mathbf{X}_n \overset{law}{\to} \mathbf{Y} \quad \Rightarrow \quad \mu_{\mathbf{X}} = \mu_{\mathbf{Y}}. \tag{4.59}$$

Proof. Since the weak limit is unique, $\mu_{\mathbf{X}_n} \to \mu_{\mathbf{X}} = \mu_{\mathbf{Y}}$. ■

Proposition 4.13. *The necessary and sufficient condition that $\mathbf{X}_n$ $(n = 1, 2, \cdots)$ converges in probability is that for any $\varepsilon > 0$,*

$$\lim_{n,m\to\infty} \mathbb{P}(\|X_n - X_m\| > \varepsilon) = 0 \tag{4.60}$$

Proof. Since

$$\mathbb{P}(\|X_n - X_m\| > 2\varepsilon) \leq \mathbb{P}(\|X_n - X\| > \varepsilon) + \mathbb{P}(\|X - X_m\| > \varepsilon) \tag{4.61}$$

the necessity is proven as $n, m \to \infty$.

By the assumption, we take a subsequence $1 = n_1 < n_2 < \cdots$ such that for $m > n_k$

$$\mathbb{P}(\|X_m - X_{n_k}\| > 2^{-k}) \leq 2^{-k}. \tag{4.62}$$

Thus we have

$$\sum_{k=1}^{\infty} \mathbb{P}(\|X_m - X_{n_k}\| > 2^{-k}) < \infty. \tag{4.63}$$

Therefore, by the Borel–Cantelli lemma,

$$\sum_{k=1}^{m} \|X_{n_{k+1}} - X_{n_k}\| \tag{4.64}$$

is a.s. convergent as $m \to \infty$, which yields that there is a limit $X := \lim_{k\to\infty} X_{n_k}$. For $n > n_k$,

$$\mathbb{P}(\|X - X_n\| > 2^{-k+1}) \leq \mathbb{P}(\|X - X_{n_k}\| > 2^{-k}) + \mathbb{P}(\|X_n - X_{n_k}\| > 2^{-k}) \leq 2^{-k+2}, \tag{4.65}$$

which shows the convergence $X_n \to X$ in probability. ■

Question 4.14. *Let $\mathbf{X}_n$, $\mathbf{X}$ be continuous stochastic processes and c be a real number. Then prove the following:*

(1) $\mathbf{X}_n \overset{a.s.}{\to} \mathbf{X} \quad \Rightarrow \quad \mathbf{X}_n \overset{pr}{\to} \mathbf{X}$

(2). *If $\mathbf{X}_n \overset{pr}{\to} \mathbf{X}$, then there exists a subsequence that converges almost surely.*

(3). $\mathbf{X}_n \overset{pr}{\to} \mathbf{X} \quad \Rightarrow \quad \mathbf{X}_n \overset{d}{\to} \mathbf{X}$

(4). $\mathbf{X}_n \overset{d}{\to} c \quad \Rightarrow \quad \mathbf{X}_n \overset{pr}{\to} c$

The following inequality yields a result about the independent sum of stochastic processes.

Lemma 4.15 (Ottaviani's inequality)**.** *Let $\xi_1, \xi_2, \cdots$ be independent C-valued random variables and $c \in \mathbb{R}$. Then we have the following inequality*

$$\mathbb{P}\left(\max_{k=1,2,\cdots,n} \|\xi_1 + \xi_2 + \cdots + \xi_k\| \geq 2c\right) \leq \frac{\mathbb{P}(\|\xi_1 + \xi_2 + \cdots + \xi_n\| \geq c)}{1 - \max_{k=1,2,\cdots,n} \mathbb{P}(\|\xi_{k+1} + \xi_{k+2} + \cdots + \xi_n\| \geq c)} \tag{4.66}$$

Proof. Let $Y_k := \|\xi_1 + \cdots + \xi_k\|$ and $Z_k := \|\xi_{k+1} + \cdots + \xi_n\|$, where $Z_n := 0$. For $c > 0$, we set

$$A_k = \{Y_1 < 2c, Y_2 < 2c, \cdots, Y_{k-1} < 2c, Y_k \geq 2c\}. \tag{4.67}$$

Since $Y_n \geq Y_k - Z_k$,

$$\{Y_k \geq 2c\} \cap \{Z_k \leq c\} \subset \{Y_n \geq c\} \tag{4.68}$$

Thus

$$A_k \cap \{Z - k \leq c\} \subset A_k \cap \{Y_n \geq c\}. \tag{4.69}$$

Since $\{Y_1, Y_2, \cdots, Y_k$ is independent of Z_k,

$$\mathbb{P}(A_k \cap \{Y_n \geq c\}) \geq \mathbb{P}(A_k \cap \{Z_k \leq c\}) = \mathbb{P}(A_k)\,\mathbb{P}(Z_k \leq c) = \mathbb{P}(A_k)\left(1 - \mathbb{P}(Z_k > c)\right) \tag{4.70}$$

Since $A_1, A_2, \cdots, A_n$ are disjoint and $\bigcup_{k=1}^n A_k = \{\max_{k=1,2,\cdots,n} Y_k \geq 2c\}$, we obtain

$$\begin{aligned}\mathbb{P}(Y_n \geq c) &\geq \mathbb{P}\left(\{\max_{k=1,2,\cdots,n} Y_k \geq 2c\} \cap \{Y_n \geq c\}\right) \\ &\geq \mathbb{P}\left(\max_{k=1,2,\cdots,n} Y_k \geq 2c\right)\left(1 - \max_{k=1,2,\cdots,n} \mathbb{P}(Z_k > c)\right),\end{aligned} \tag{4.71}$$

which implies the desired inequality. ■

Proposition 4.16. *Let $\xi_1, \xi_2, \cdots$ be independent C-valued random variables and set*

$$\mathbf{X}_n = \sum_{k=1}^n \xi_k \tag{4.72}$$

Then

$$\mathbf{X}_n \overset{pr}{\to} \mathbf{X} \quad \Rightarrow \quad \mathbf{X}_n \overset{a.s.}{\to} \mathbf{X} \tag{4.73}$$

Proof. Since $\mathbf{X}_n$ converges in probability, there is an $N = N(j)$ such that

$$\sup_{N \leq m < n} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_m\| > 2^{-j}) < 2^{-j}. \tag{4.74}$$

Let $\eta_1 = \xi_{N+1}, \eta_2 = \xi_{N+2}, \cdots$. Then, by Lemma 4.15, we have

$$\mathbb{P}\left(\max_{k=1,2,\cdots,n} \|\mathbf{X}_{N+k} - \mathbf{X}_N\| > 2^{-j+1}\right) < 2^{-j+1}. \tag{4.75}$$

As $n \to \infty$, we obtain

$$\mathbb{P}\left(\sup_{k \geq 1} \|\mathbf{X}_{N+k} - \mathbf{X}_N\| > 2^{-j+1}\right) < 2^{-j+1} \tag{4.76}$$

Thus, $\sum_j \mathbb{P}\left(\sup_{k\geq 1} \|\mathbf{X}_{N+k} - \mathbf{X}_N\| > 2^{-j+1}\right) < \infty$. Therefore, by the Borel–Cantelli lemma, for almost all ω, there is $j_0(\omega)$ such that for $j \geq j_0(\omega)$,

$$\sup_{k \geq 1} \|\mathbf{X}_{N(j)+k}(\omega) - \mathbf{X}_{N(j)}(\omega)\| > 2^{-j+1}, \tag{4.77}$$

which implies that $X_n(\omega)$ is a convergent sequence on C. ■

We skip the proof of following proposition, but it yields the totally bounded condition for the family of distributions on C.

Proposition 4.17. *Let $\mathbf{X}_n$ $(n = 1, 2, \cdots)$ be continuous stochastic process satisfying*

$$M := \sup_n \max_{j=1,2,\cdots,d} \mathbb{E}[X_{n,j}^2(0)] < \infty \tag{4.78}$$

and for positive numbers a, γ, δ,

$$\sup_n \max_{j=1,2,\cdots,d} \mathbb{E}[|X_{n,j}(t) - X_{n,j}(s)|^\gamma] \leq a\,|t-s|^{1+\delta} \tag{4.79}$$

Then we have

(1). *arbitrary subsequence $\mathbf{X}_{n_k}$ $(k = 1, 2, \cdots)$ has a subsequence $\mathbf{X}_{n_{k_j}}$ $(j = 1, 2, \cdots)$ converging in law*

(2). *If all subsequence converging in law have the same limit, then $\mathbf{X}_n$ $(n = 1, 2, \cdots)$ converges also in law.*

In the next section, we see the Levy's construction of Brownian motion. The proposition above plays an important role when we consider the Fourier series approximation for Brownian motion. The construction of Brownian motion by Fourier series was done by N. Wiener and it is called *Wiener expansion*.

4.3 Examples of continuous time stochastic process

4.3.1 Poisson process

Let $S_0 = 0$ and $S_1 < S_2 < \cdots$ be the sequence of time when an event A occurs. Suppose that the increments $\eta_n = S_n - S_{n-1}$ $(n = 1, 2, \cdots)$ are independent and identically distributed by exponential distribution with parameter $\lambda > 0$. Then

$$S_n = \sum_{k=1}^{n} \eta_k \tag{4.80}$$

$\{S_n\}_{n=0,1,2,\cdots}$ is a random walk. Let N_t be the number of appearance of the event A until time t:

$$N_t = \min\{k \geq 0 \colon S_{k+1} > t\} = \sum_{n=1}^{\infty} \mathbb{1}_{[0,t]}(S_n) \tag{4.81}$$

Then the path $N_\cdot(\omega)$ is a right-continuous step function at $t = S_n(\omega)$ with jump width 1. We treat Poisson process again in the latter section.

Question 4.18. Prove that the distribution of N_t is Poison distribution.

4.3.2 Renewal process

The renewal process is an extension of Poisson process. Let $S_0 = 0$ and $S_1 < S_2 < \cdots$ be the sequence of time when an event A occurs. Suppose that the increments $\eta_n = S_n - S_{n-1}$ $(n = 1, 2, \cdots)$ are i.i.d., not only exponential distribution. We denote

$$\mathbb{P}(\eta_n \leq t) =: F(t), \qquad F(0) = \mathbb{P}(\eta_n \leq 0) = 0 \tag{4.82}$$

Let T_n be the n-th renewal. We define the number of renewals until time t by

$$N(t) = \max\{n \colon T_n \leq t\} \tag{4.83}$$

We call $N(t)$ the **renewal process**. Then we have

$$\{N(t) \geq n\} = \{T_n \leq t\} \tag{4.84}$$

Now we consider the average of renewals until time T as

$$\mathbb{E}[T] = \int_0^\infty \mathbb{E}[\mathbb{1}_T(t)]\, dt = \int_0^\infty \mathbb{P}(T > t)\, dt = \sum_{m=1}^{\infty} \mathbb{P}(T \geq m) \tag{4.85}$$

which implies that

$$\mathbb{E}[N(t)] = \int_0^\infty \mathbb{P}(N(t) \geq t)\, dt = \int_0^\infty \mathbb{P}(T_n \leq t)\, dt \tag{4.86}$$

Lemma 4.19 (Wald's equation)**.** *Let τ be a stopping time with $\mathbb{E}[\tau] < \infty$ and $\mu = \mathbb{E}[\eta_n]$. Then we have*

$$\mathbb{E}[S_\tau] = \mu\, \mathbb{E}[\tau] \tag{4.87}$$

Proof.

$$S_\tau = \sum_{m=1}^{\infty} \eta_m \, \mathbb{1}_\tau(m) \tag{4.88}$$

Then

$$\mathbb{E}[S_\tau] = \sum_{m=1}^{\infty} \mathbb{E}[\eta_m] \, \mathbb{E}[\mathbb{1}_\tau(m)] = \mu \sum_{m=1}^{\infty} \mathbb{P}(\tau \geq m) = \mu \, \mathbb{E}[\tau] \tag{4.89}$$

■

By (4.86),

$$\mathbb{E}[N(t)] = \sum_{m=1}^{\infty} \mathbb{P}(N(t) \geq m) = \sum_{m=1}^{\infty} \mathbb{P}(T_m \leq t) \tag{4.90}$$

Let $\tau = \min\{n \colon T_\tau > t\} = N(t) + 1$. Then τ is the stopping time. By Wald's equation,

$$\mathbb{E}[T_{N(t)+1}] = \mu \, \mathbb{E}[N(t) + 1] \tag{4.91}$$

From (4.84),

$$T_{N(t)+1} > t \tag{4.92}$$

$$\frac{t}{\mu} \leq \mathbb{E}[N(t)] + 1 \tag{4.93}$$

Question 4.20. If $\eta_i < M$ for all i, then prove that

$$\frac{t + M}{\mu} \geq \mathbb{E}[N(t)] + 1 \tag{4.94}$$

If η is not bounded, we consider *truncation* by M, i.e.

$$\eta_i^{(M)} := \min\{\eta_i, M\} \tag{4.95}$$

Then the truncated counted process $N^{(M)}(t)$ by $\eta^{(M)}(t)$ satisfies

$$N^{(M)}(t) \geq N(t) \tag{4.96}$$

and $\mathbb{E}[\eta_i^{(M)}] \to \mu$ as $M \to \infty$.

Proposition 4.21. *Let $\mu = \mathbb{E}[\eta_n]$. Then as $t \to \infty$, we have*

$$\frac{N(t)}{t} \to \frac{1}{\mu} \tag{4.97}$$

Proof. By the definition of renewal process,

$$T_{N(t)} \leq t < T_{N(t)+1} \tag{4.98}$$

Then we have

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)+1} \frac{N(t)+1}{N(t)} \tag{4.99}$$

Since $N(t) \to \infty$ as $t \to \infty$, the strong law of large numbers, we have

$$\frac{S_{N(t)}}{N(t)} \to \mathbb{E}[\eta] = \mu \tag{4.100}$$

By the squeeze theorem, we conclude the proof. ■

Remark 4.22. By (4.93) and (4.94), we have

$$\frac{\mathbb{E}[N(t)]}{t} \to \frac{1}{\mu} \qquad t \to \infty \tag{4.101}$$

4.3.3 Gaussian process

For $X = (X_1, X_2, \cdots, X_d)$, $a \in \mathbb{R}^d$ and d-dimensional symmetric positive semi-definite matrix $V = (v_{ij})_{i,j=1,2,\cdots,d}$, X is called d-dimensional **Gaussian** random variable when the characteristic function $\varphi_X(z)$ is expressed by

$$\varphi_X(z) = \exp\left(i\langle z, a\rangle - \frac{1}{2} z V z^{\mathsf{T}}\right), z = (z_1, z_2, \cdots, z_d) \in \mathbb{R}^d, \tag{4.102}$$

where $\langle z, a\rangle = \sum_{i=1}^d z_i a_i$ and z^{T} is a transpose of z. By differentiating the characteristic function $\varphi_X(z)$,

$$a_k = \mathbb{E}[X_k], \qquad v_{ij} = \mathrm{Cov}(X_i, X_j) \tag{4.103}$$

a is called **mean vector**, V is a **covariance matrix** and we denote by $N(a, V)$ the Gaussian.

Definition 4.23 (Gaussian process)**.** For any n and $t_1, t_2, \cdots, t_n$, if the distribution of $X = (X_{t_1}, X_{t_2}, \cdots, X_{t_n})$ is dn-dimensional Gaussian distribution, then d-dimensional stochastic process $\{X_t\}_{t\in I}$ is called (d-dimensional) **Gaussian process**.

Definition 4.24 (Independency)**.** Let $\mathbf{X}^{(k)} = \{X_t^{(k)}\}_{t\in I_k}$ $(k = 1, 2, \cdots, n)$ be a a d_k-dimensional stochastic process. For any m_k, $t_{k,1}, t_{k,2}, \cdots, t_{k,m_k}$, if n multi-dimensional random variables

$$(X_{t_{k,1}}^{(k)}, X_{t_{k,2}}^{(k)}, \cdots, X_{t_{k,m_k}}^{(k)}) \qquad k = 1, 2, \cdots, n \tag{4.104}$$

are independent, then $\mathbf{X}^{(1)}$, $\mathbf{X}^{(2)}$, $\cdots$, $\mathbf{X}^{(n)}$ are called **independent**.

We note here that when $I_k = \{0\}$, $\mathbf{X}^{(k)}$ is a just d_k-dimensional random variable, not a stochastic process. Thus Definition 4.24 means the definition of independence for both random variables and stochastic processes.

Proposition 4.25. *Let $Y = (Y_1, Y_2, \cdots, Y_d)$ be a d-dimensional Gaussian random variables and $\mathbf{X} = \{X_t\}_{t\in I}$ be a Gaussian process. Then the necessary and sufficient condition for independence of Y and $\mathbf{X}$ is for any $t \in I$,*

$$\mathrm{Cov}(Y_k, X_t) = 0 \qquad k = 1, 2, \cdots d \tag{4.105}$$

Proof. The proof is exactly the same as Proposition in Note_1. The necessary and sufficient condition for the independence of Y and $(X_{t_1}, X_{t_2}, \cdots, X_{t_n})$ is

$$\mathrm{Cov}(Y_k, X_{t_j}) = 0 \qquad k = 1, 2, \cdots, d, \quad j = 1, 2, \cdots, n \tag{4.106}$$

■

4.3.4 Stationary process

Definition 4.26 (Stationary process)**.** d-dimensional stochastic process $\{X_t\}_{t\geq 0}$ or $\{X_t\}_{t\in\mathbb{R}}$ is called (d-dimensional) **stationary process** when the distribution is invariant with respect to the time shift, i.e. for any n, $t_1, t_2, \cdots, t_n$ and h,

$$\text{distribution of } (X_{t_1+h}, X_{t_2+h}, \cdots, X_{t_n+h}) = \text{distribution of } (X_{t_1}, X_{t_2}, \cdots, X_{t_n}) \tag{4.107}$$

Let $\mathbf{X} = (X_t)_{t\geq 0}$ be a Gaussian process. Then the distribution of $\mathbf{X}$ is determined by the mean $a(t) := \mathbb{E}[X_t]$ and $\Gamma(t, s) := \mathrm{Cov}(X_t, X_s)$. If $a(t)$ is a constant and $\Gamma(t, s)$ is a function of $t - s$, then the Gaussian process becomes a stationary process.

On the other hand, let $\mathbf{X} = (X_t)_{t\in\mathbb{R}}$ be a stationary process. Then $\mathbb{E}[X_t]$ is irrelevant to t, and $\mathrm{Cov}(X_t, X_s)$ is a function of $t-s$. The covariance function $r(t) := \mathrm{Cov}(X_t, X_0)$ is continuous by the assumption of square mean continuous, and satisfies the following

- $r(-t) = r(t)$
- $|r(t)| \le r(0)$
- for any positive integer N and real numbers $t_1, t_2, \cdots, t_N$, $x_1, x_2, \cdots, x_N$,

$$\sum_{k,j=1}^{N} r(t_k - t_j)\, x_k\, x_J \ge 0 \tag{4.108}$$

From those properties, the function $r(\cdot)$ have **spectral expression**

$$r(t) = \int_0^\infty \cos(st)\, dG(s) = \frac{1}{2}\int_{-\infty}^{\infty} \exp\big(ist\big)\, d\widetilde{G}(s), \tag{4.109}$$

where G is an increasing bounded function with $G(0) = 0$, and is called **spectral density**, which is determined uniquely by $r(\cdot)$. When G is differentiable, the derivative G' is called **spectral density function**.

A Review of Calculus

Continuity The continuity of f at x_0 is defined as

$$f(x_0) = \lim_{x\to x_0} f(x) \tag{A.110}$$

To be more precise, the left and right limit coincide:

$$L^- := \lim_{x\to x_0^-} f(x), \qquad L^+ := \lim_{x\to x_0^+} f(x) \tag{A.111}$$

We classify the discontinuity

- if $L^- = L^+$ but $f(x_0) \ne \lim_{x\to x_0} f(x)$, then f has **removable discontinuity** at $x = x_0$.
- if $L^- \ne L^+$, then f has **jump discontinuity** at $x = x_0$.
- if each of L^- or L^+ is not finite (the limit does not exist), then x_0 is called **essential discontinuity** of f.

Combining removable and jump discontinuities, we call **discontinuity of the first kind**. The essential discontinuity is called the **dicontinuity of the second kind**. When $L^+ \ne f(x_0)$ ($L^- \ne f(x_0)$), f is called right (left) continuous.